

RESEARCH ARTICLE

WILEY

Real-time implementation of adaptive PV-integrated SAPF to enhance power quality

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Summary

This paper presents the design and implementation of a photovoltaic-integrated shunt active power filter (SAPF) to improve the power quality and to generate clean power. The system uses adaptive neuro-fuzzy inference system (ANFIS)-based maximum power point tracking and control of synchronous reference frame theory-based SAPF. Various control schemes are implemented in MATLAB and then validated in real-time using FPGA-based computation engine of OPAL-RT 4510. Control techniques built around the artificial neural network, fuzzy logic control, and ANFIS are compared for balanced and unbalanced loads on parameters like total losses with/without compensation, voltage drop, power factor, and total harmonic distortion.

KEYWORDS

adaptive neuro-fuzzy, maximum power point tracking, power quality, shunt active power filter, synchronous reference frame theory

1 | INTRODUCTION

Power quality problems in the present context has seen tremendous rise because of the increased penetration of nonlinear loads. Moreover, the number of electric vehicles (EVs) on the road has increased and its charging is creating power quality problems for the grid. Poor quality power supply results in substandard performance of the equipment, therefore, harmonics elimination becomes mandatory in power systems. A shunt active power filter (SAPF) provides the compensating current to nullify the alternating current (AC) side current harmonics. The reference current is extracted while controlling direct current (DC)-link voltage when compensation procedure is applied to voltage source inverter (VSI). The conventional controllers like proportional integral (PI) and PI derivative (PID) controller yield lower performance during varying load conditions. Artificial intelligence (AI) techniques can easily handle fault tolerance and nonlinearities therefore used aggressively to solve complex problems.¹ The selection of control method plays a significant

List of symbols and abbreviations: Δ_{iL} , ripple filter inductance; %, percentage; C_{dc} , DC-link capacitor; D , duty cycle; i_c , compensating current; i_{La} , i_{Lb} , i_{Lc} , load currents; i_{sa}^* , i_{sb}^* , i_{sc}^* , reference source currents; i_{sa} , i_{sb} , i_{sc} , actual source current; i_d , i_q , direct and quadrature axis current; k , overloading factor; m_i , modulation index; n , nth harmonic order; V_{sa} , V_{sb} , V_{sc} , source voltages; V_{dc} , dc link voltage; V_{dc}^* , reference dc link voltage; AI, artificial intelligence; ADALINE, adaptive linear neuron; ANFIS, adaptive neuro fuzzy inference system; ANN, artificial neural network; BP, back propagation; CPV, capacitance of PV array; DSTATCOM, distribution static compensator; EVs, electric vehicles; FIS, fuzzy inference system; FLC, fuzzy logic control; FPGA, field programmable gate array; HIL, hardware in the loop; I&C, incremental and conductance; IRPT, instantaneous reactive power theory; MF, membership function; MNN, multilayer neural network; MPPT, maximum power point tracking; PCC, point of common coupling; PID, proportional integral derivative; P&O, perturb & observe; PLL, phase lock loop; PV, photovoltaic; RTDSH, real-time digital simulation hardware; SAPF, shunt active power filter; SRF, synchronous reference frame; THD, total harmonic distortion; UPQC, unified power quality conditioner; VMPP, PV array voltage at maximum power point; VPV, PV array voltage; VSI, voltage source inverter.

role in the performance of SAPF. The following section briefly presents the literature review on the usage of AI techniques in SAPF.

The adaptive linear neuron (ADALINE) and feed-forward multilayer neural network (MNN) methods to extract harmonic current in SAPF have been demonstrated.^{2,3} The radial basis function-neural network has been used to reduce the total harmonic distortion (THD) in the SAPF system.⁴ The current control of SAPF using artificial neural network (ANN)-based controller have been designed and trained offline, utilizing the traditional PI controller.^{5,6} The main purpose of this controller is to control the DC-link voltage, eliminate the harmonic component, and provide fast transient response. The ANN-based grid synchronization method issued for phase-locking algorithm using ADALINE has been implemented.^{7,8} ADALINE-based phase lock loop (PLL) is unsuitable for varying load conditions. Control of the DC-link voltage and minimization of THD using PI and fuzzy logic-based control strategy is presented.⁹⁻¹¹ The adaptive neuro-fuzzy inference system (ANFIS) with multi-criteria decision-making have been presented to handle the uncertainties and for precise and reliable results. The SAPF has been effectively controlled to minimize at the same time the THD, reactive power, and the current imbalance of nonlinear load with the generated renewable power injection to the grid. The ANFIS-based dynamic APF to enhance the power quality of grid at source side with effective integration of renewable sources have been presented.¹²

As the energy demand is increasing at a rapid pace and our conventional resources are limited, thus renewable energy may serve as a possible solution to this problem. Solar PV is one of the major contributors to renewable power generations as it is freely available and eco-friendly to the environment. The power electronic converters have been used as an interface to PV array and are connected with the grid. The voltage and current characteristics of PV array changes with irradiation and temperature. To track the maximum power from a PV module, maximum power point tracking (MPPT) algorithms have been used. Numerous MPPT strategies have been created and actualized such as perturb and observe (P&O), incremental and conductance (I&C), fractional open circuit voltage and fractional short circuit current, etc.^{13,14} An ANN-based method to track the maximum power from a photovoltaic (PV) module for variable cell temperature and radiation was introduced.¹³ A two-stage MPPT controller using ANN for PV systems under fluctuating solar irradiation and temperature was designed.¹⁵ The ANN-based MPPT methods have faster tracking speed, no oscillations around the MPP, high tracking efficiency, and can be used for all types of power converters. The performance of traditional MPPT algorithms is poor as compared with AI-based methodology, particularly in partial shading and fast changing climatic conditions. Nonetheless, AI-based methodologies have high-speed response and impose less computational burden.¹⁶ The ANFIS-based MPPT under fast changing irradiation conditions has been described.¹⁷ The ANFIS-based controller for quick changing solar irradiance and cell temperature has also been introduced.¹⁸ It has been concluded that a precise and fast response can be obtained using ANFIS controller. Denoising-based wavelet MPPT algorithm has been implemented in Chikh and Chandra,¹⁹ which minimized the sensor requirements because of its fast tracking capabilities for all environmental conditions.

The PV-integrated SAPF is a growing technology in the field of power systems and it is very useful in remote locations where transmission networks are unavailable. During the day, the PV-integrated SAPF, not only powers the load but also improves the power quality however at the night time, it only improves the power quality.²⁰ The operation of PV-integrated SAPF consisting of DC-DC boost converter have been demonstrated.²¹ They have used P&O algorithm to extract the highest power and applied the instantaneous reactive power theory (IRPT) to mitigate harmonics. A PV-integrated SAPF have been demonstrated using I&C method to extract MPP and IRPT method to control SAPF.^{22,23}

The IRPT strategy has been inefficient for dynamic load changes and has slower response and large computational burden as compared with the synchronous reference frame (SRF) theory. PV-based distribution static compensator (DSTATCOM) to improve power quality using SRF theory has been presented.^{24,25} It provides better control for reactive power compensation, minimizes the voltage variations, and yields better utilization of solar energy. Wavelet-based strategy has been implemented to minimize the harmonic distortion, voltage sag, and swell.²⁶ A single-phase modified SRF theory based multilevel converter for power quality enhancement has also been demonstrated.²⁷ In Renukadevi et al,²⁸ an infinite level VSI, which acts as a DSTATCOM, has been applied to reduce the THD. A 3-level neutral point clamped inverter-based single-phase SAPF has been utilized to minimize the THD and maintain DC-link voltage.²⁹ PV-integrated unified power quality conditioner (UPQC) has been proposed in Patel et al,³⁰ which has various advantages like protecting critical loads from grid side disruption, enhancing power quality, and increasing the fault ride-through capacity of the converter during transients. PV-integrated DSTATCOM using adaptive techniques has been implemented to improve the power quality problem and to transfer the power to the load, decreasing the generation demand through traditional sources.³¹ PV-based DC-DC boost converter fed VSI using IRPT-driven control strategy for power quality improvement has been described.³² The PV-integrated DSTATCOM performs dual functions of

improving power quality and also powering the load, which in turn reduces the generation requirement through conventional sources.

A number of solutions like, STATCOM, DSTATCOM, and SAPF have been used to mitigate the power quality issues produced by charging of EVs.³³ Fuzzy controller has been employed to address the problems of power quality originating from charging of EVs and improve the efficiency of the network.³⁴

The majority of the work available in the literature concentrates on MPPT operation in PV array, design of inverter topologies, and performance of inverter in grid integration. This paper presents a novel method wherein it applies ANFIS controller for the complete PV-integrated SAPF system. The results are compared with various other techniques, and it has been found that it effectively tracks the MPP in smaller duration with lesser fluctuations.

The proposed system design is simulated in MATLAB/Simulink environment and validated in real-time using real-time digital simulation hardware (RTDSH), OPAL-RT 4510. This system has double stage topology. In first stage, DC-DC boost converter steps up the voltage, and in second stage, DC-AC converter (ie, VSI, which works as SAPF) is controlled by reference current generation technique. The major objectives and contributions of this paper are:

1. To design ANFIS-based controller to harvest highest extent of power from PV array (MPPT).
2. To design ANFIS-based adaptive technique to improve the performance of PV-SAPF under steady state as well as dynamic load conditions.
3. To minimize the THD of source current and to attain purely balanced and sinusoidal source current.
4. To evaluate the performance using MATLAB/Simulink and its validation in real-time using OPAL-RT 4510 platform for different loads.
5. To compare control techniques for balanced and unbalanced nonlinear loads, with/without compensation on parameters like total losses, voltage drop, power factor, and THD.
6. To arrive at cost benefits analysis for various control techniques.

The remaining part of the paper is divided into six sections wherein Section 2 presents the proposed system configuration of PV-integrated SAPF and Section 3 describes the control strategies of SAPF. Section 4 presents the simulation results and Section 5 elaborates the real-time results and comparative analysis. Finally, conclusions drawn are presented in Section 6.

2 | PROPOSED SYSTEM CONFIGURATION OF PV-INTEGRATED SHUNT ACTIVE POWER FILTER

The power circuit of the proposed PV-SAPF system is shown in Figure 1. The boost converter is connected with VSI at the point of common coupling (PCC) with nonlinear/reactive loads. The three-phase grid tied VSI has six insulated-gate bipolar transistors (IGBT) as switching devices, which act as power controller between DC-link and utility. The system

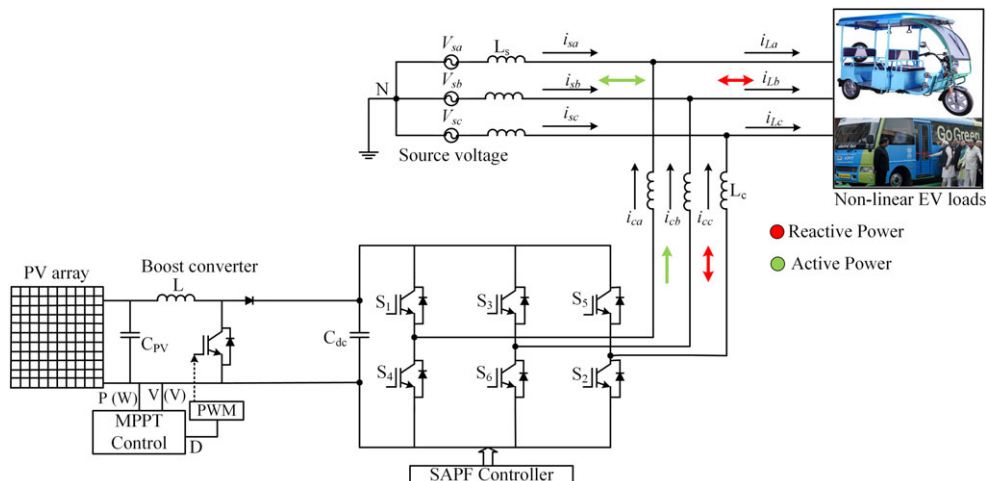


FIGURE 1 Power circuit of photovoltaic (PV)-shunt active power filter (SAPF)

performance has been optimized using ANFIS-based control algorithms. Figure 2 shows the flowchart for working of a PV-integrated SAPF. To design a required SAPF, the values of governing parameters need to be selected and are presented as follows:

Three-phase grid voltage (V_s) 400 V, frequency (f) 50 Hz, load current (i_L) 150 A, and supply inductance 1 mH. The EVs and other nonlinear loads have been represented as a combination of three-phase diode rectifier with RL. The load resistances (R) are 20 Ω , 15 Ω , and 10 Ω , and the inductances (L) are 30 mH, 20 mH, and 10 mH, respectively. The balanced load is of 10 kW, and the unbalanced load in the phases is of 10 kW, 9 kW, and 11 kW, respectively. The PV-SAPF has been designed based on the analysis available in a previous study,³⁵ and their parameters are mentioned in Table 1 below:

2.1 | Maximum power point tracking (MPPT)

The PV has nonlinear V-I characteristics that changes with solar irradiance and temperature as shown in Figure 3. Therefore, it is imperative to track the MPP. MPP tracking has been implemented using DC-DC converters to achieve maximum efficiency of PV array. Boost and buck converters are mostly used in DC-DC converter topologies. However, boost converter shows advantages over buck converter because of the blocking diodes that can prevent reverse current flow into PV array. PV array parameters V_{oc} and I_{sc} and environmental parameters irradiance and temperature form the input matrix and the output is in terms of duty cycle corresponding to MPP that is used to drive the boost converter. The PWM module generates signal to control the duty cycle of the IGBT switch. Here, ANFIS control technique is used for MPPT. The inputs of the proposed MPPT controller are the power variation (ΔP) and voltage variation (ΔV), and output, duty cycle variation (ΔD).

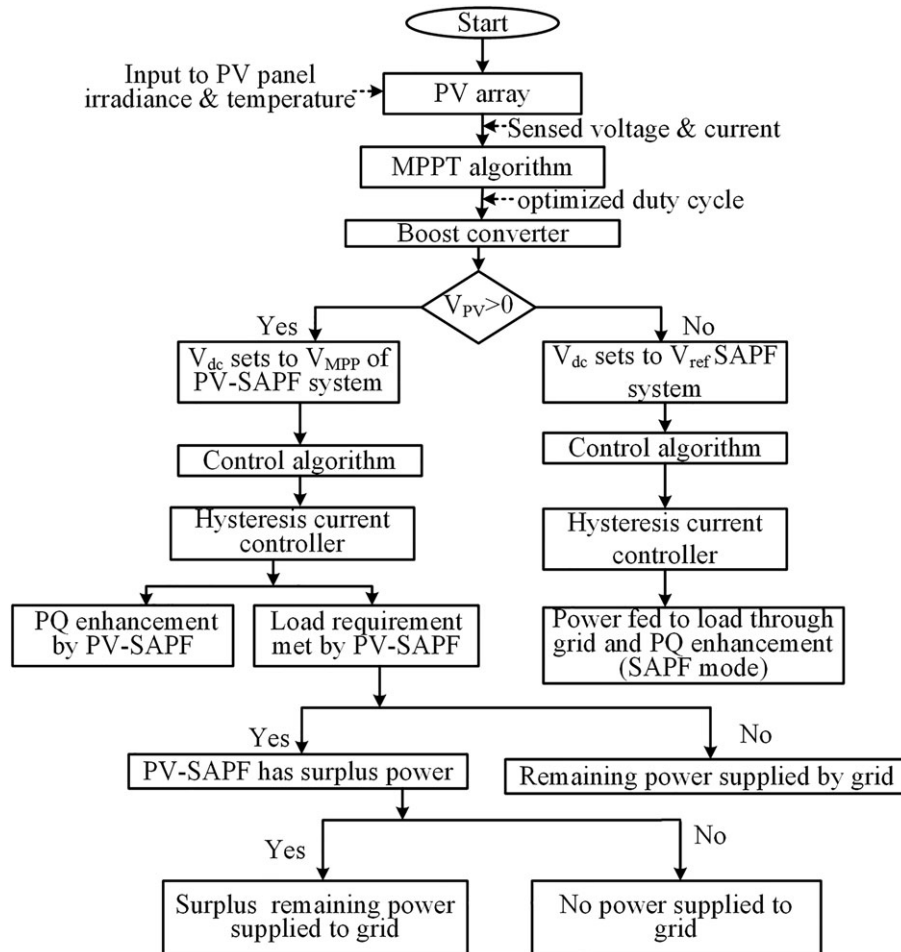
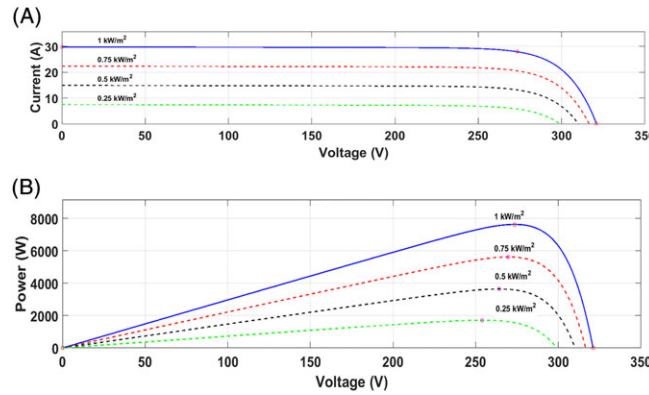


FIGURE 2 Flowchart showing operation of photovoltaic (PV)-integrated shunt active power filter (SAPF)

TABLE 1 Design parameters of photovoltaic (PV)-integrated shunt active power filter (SAPF)³⁵

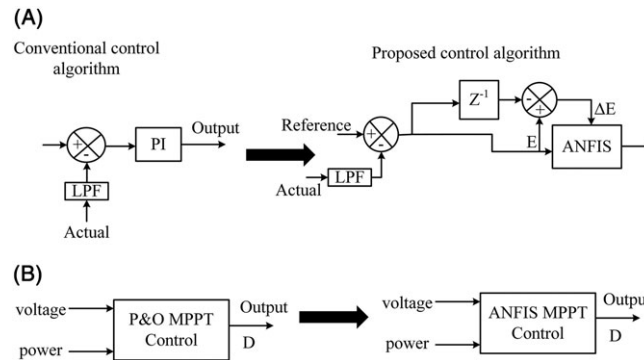
Description	Parameter
Design of boost converter	$L = \frac{V_{PV}(V_{dc} - V_{PV})}{\Delta i_L f V_{dc}} = 3.75 \text{ mH}$
Output capacitance of PV array	$C_{PV} = \frac{DV_{PV}}{4f^2 \Delta L_{PV} \Delta C_{PV}} = 585 \mu\text{F}$
DC-link voltage of SAPF	$V_{dc} = \frac{2\sqrt{2}V_s}{m_i} = 750 \text{ V}$
Compensating inductance of SAPF	$L_f = \frac{\sqrt{3}m_i V_{dc}}{6kf\Delta} I_f = 4 \text{ mH}$
DC-link capacitor	$C_{dc} = \frac{3V_s k i_c \Delta t}{0.5(V_{dc}^2 - V_{dc \text{ ref}}^2)} = 5750 \mu\text{F}$

**FIGURE 3** A, Voltage and current characteristics of photovoltaic (PV) array. B, PV array characteristics with different irradiance

2.2 | ANFIS-based control algorithm

Figure 4. A, Shows the traditional PI and the proposed ANFIS controller. B, Shows the traditional P&O and the proposed ANFIS-MPPT controller to control the SAPF and MPPT systems, respectively.

The key advantage of the soft computing tools is that a mathematical model is not a prerequisite. The model of the considered system can be constructed with the available input-output data of the system. The inspiring features of the NN are its ability to learn and adapt, whereas the capability of a fuzzy system is to consider imprecision and prevailing uncertainties. To exploit the advantages of both methods, the ANFIS algorithm comprising of both the NN and the fuzzy logic is adopted for the proposed work.

**FIGURE 4** A, Conventional and proposed strategy of shunt active power filter (SAPF). B, conventional and proposed strategy of maximum power point tracking (MPPT)

The proposed system deals with highly uncertain climatic condition of solar irradiation, temperature, and load variations. Concurrently, the system should also be adaptive and responsive to these changes. Therefore, application of ANFIS is appropriate for the proposed system. Further, the literature survey establishes that ANFIS results in an accurate modeling of the system, provided a good data bank is available. This is an icing on the cake for the system under consideration as the authors can generate the data set. As expected, the ANFIS-based MPPT and SAPF controller provides lesser voltage variation during dynamically load changes, lesser settling time, improved power factor, and diminishes the THD as compared with conventional controllers. The real-time verification ensures that this can be implemented practically.

ANFIS is a hybrid of ANN and FLC, which enjoys benefits of both. A dynamic and parallel processing system estimates input-output functions. A fuzzy method adaptively deduces and alters its fuzzy relationship from representative mathematical samples. The NN, on the other hand, can erratically create and refine fuzzy rules from the training data. The fuzzy sets are thought to be constructive in the intelligent field and in dealing with higher processing.

The performance of FLC depends on the rule basis, number, and forms of membership functions (MFs). In this work, various MFs like triangular, trapezoidal, and Gaussian are evaluated. Out of these, triangular is chosen here as it offers minimum THD in grid current and less computational burden. The authors have tested the system with fuzzy MF rules like 3×3 , 5×5 , and 7×7 , but they all yield the same THD and settling time. Therefore, 3×3 rules are chosen here because of lesser computational burden. The fuzzy input, along with triangular MF, is fed into the NN block. The NN block consists of a rule base, which is connected to the fuzzy inference system (FIS). Here, back propagation (BP) algorithm is used to train the FIS, and the BP learning algorithm has been used for training feedforward NNs. The most commonly used FIS are Mamdani and Sugeno. In this work, the Sugeno-type FIS is used because it is computationally efficient than the Mamdani type, and can be trained by real data as opposed to the latter, which is more susceptible on an expert system. Also, the output MF of the Sugeno system is either linear or constant.³⁶

Takagi Sugeno FIS is executed here using ANFIS and has a five-layered structure as shown in Figure 5. Fuzzification of the input variable has been done in the first hidden layer and T-standard operators to compute the rule are conveyed to the second hidden layer. Third and fourth hidden layers standardize the rule strengths and determine consequent parameters of the rule, respectively. The fifth layer computes the overall input by summing all the incoming signals.

The ANFIS architecture is an adaptive feedforward network structure. Every node plays out a specific function on input signals as well as on the set of parameters of that node. Some of the nodes are adaptive, which means that the node parameters are dependent on the other nodes; whereas others are fixed nodes, where the node parameters are independent. The ANFIS needs a set of input-output data as it provides a set of MF for mapping the input data to

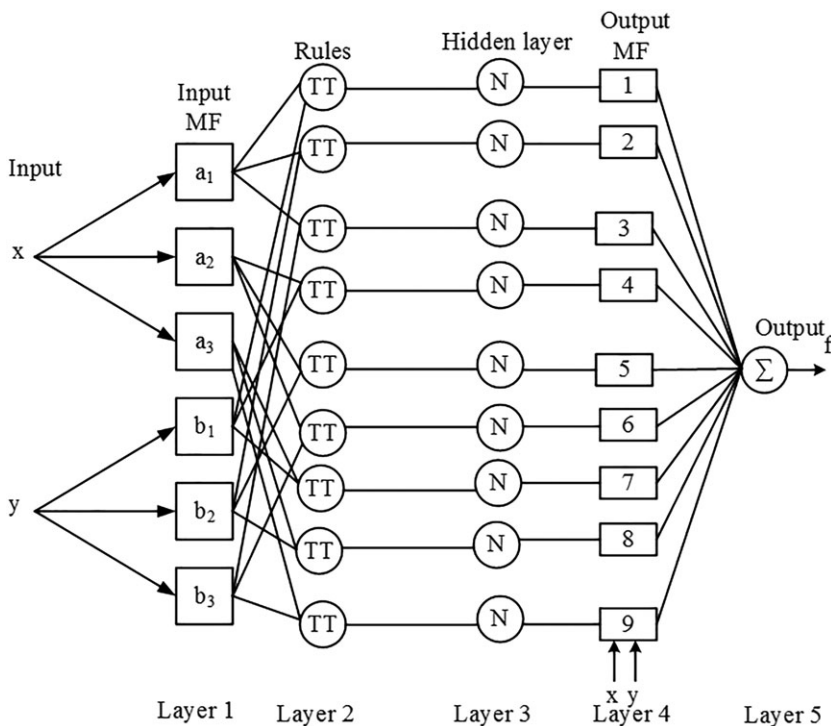


FIGURE 5 Structure of adaptive neuro-fuzzy inference system (ANFIS) controller

the output. The MF chosen should be able to minimize the error between the actual output data and ANFIS input sketched output data. The parameters connected with the MF will change within the learning process.

Layer 1: this layer contains two inputs, one as error and other as change in error represented by x and y respectively. It is commonly known as the fuzzification layer.

Layer 2: every node of the 2^n (the number of fuzzy if-then rules) is fixed and labeled as $\prod$. It multiplies the input signal and forwards and outputs the product as given by Equation 1

$$w_k = \mu_{ai}(x) * \mu_{bi}(y), \quad (1)$$

where $k = 1, 2, 3 \dots 9$. Output of each node represents the firing strength $\bar{w}_i (i = 1 \text{ to } 2^n)$ of the i th fuzzy if-then rule given by Equation 2 as

$$\bar{w}_i = \frac{w_i}{\sum_{i=1}^{2n} w_i}. \quad (2)$$

Layer 3: each node in this layer circle calculates the ratio of the individual rule's firing strength to the sum of all rules given by Equation 3 as

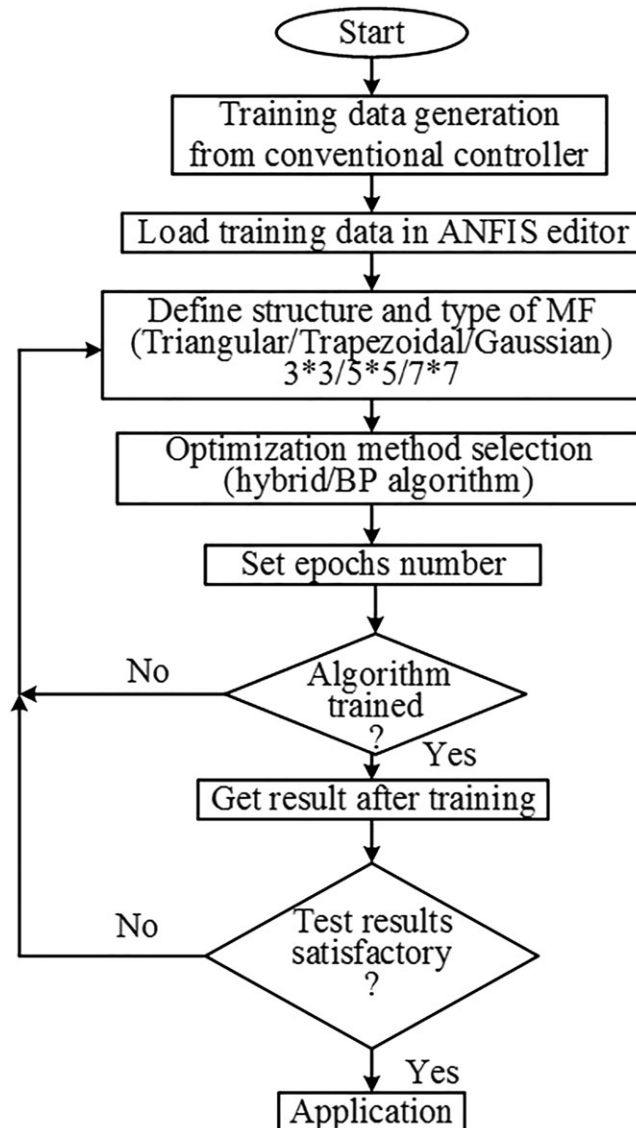
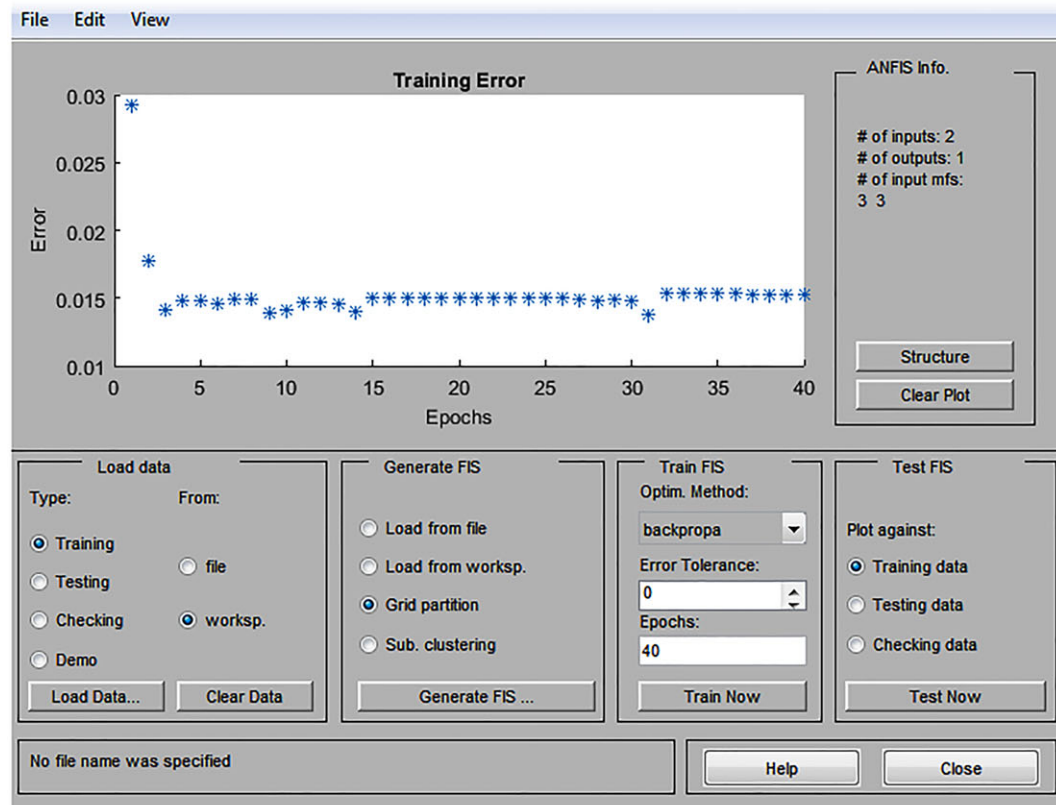


FIGURE 6 Flowchart showing the operation of adaptive neuro-fuzzy inference system (ANFIS) algorithm

(A)



(B)

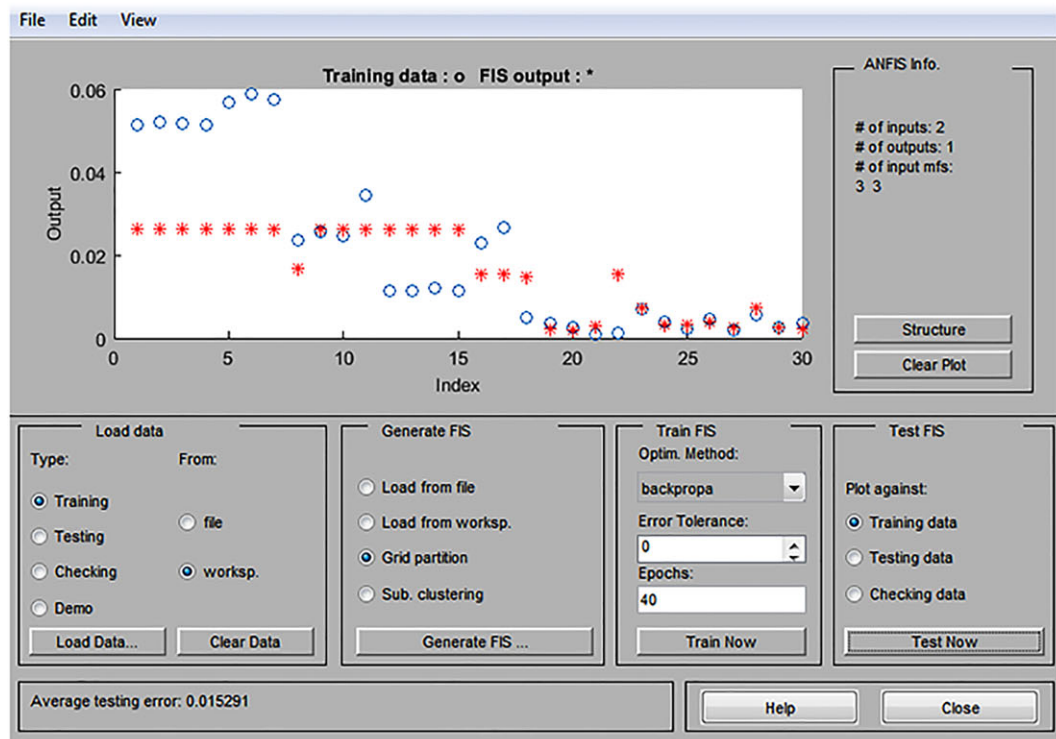


FIGURE 7 A, Epochs and actual training data error in shunt active power filter (SAPF) controller. B, Epochs and test data error in SAPF controller. Predicted output is indicated by red stars, and the test data is represented by blue circles. C, Epochs and actual training data error in maximum power point tracking (MPPT) controller. D, Epochs and test data error in MPPT controller. Predicted output is indicated by red stars, and the test data is represented by blue circles

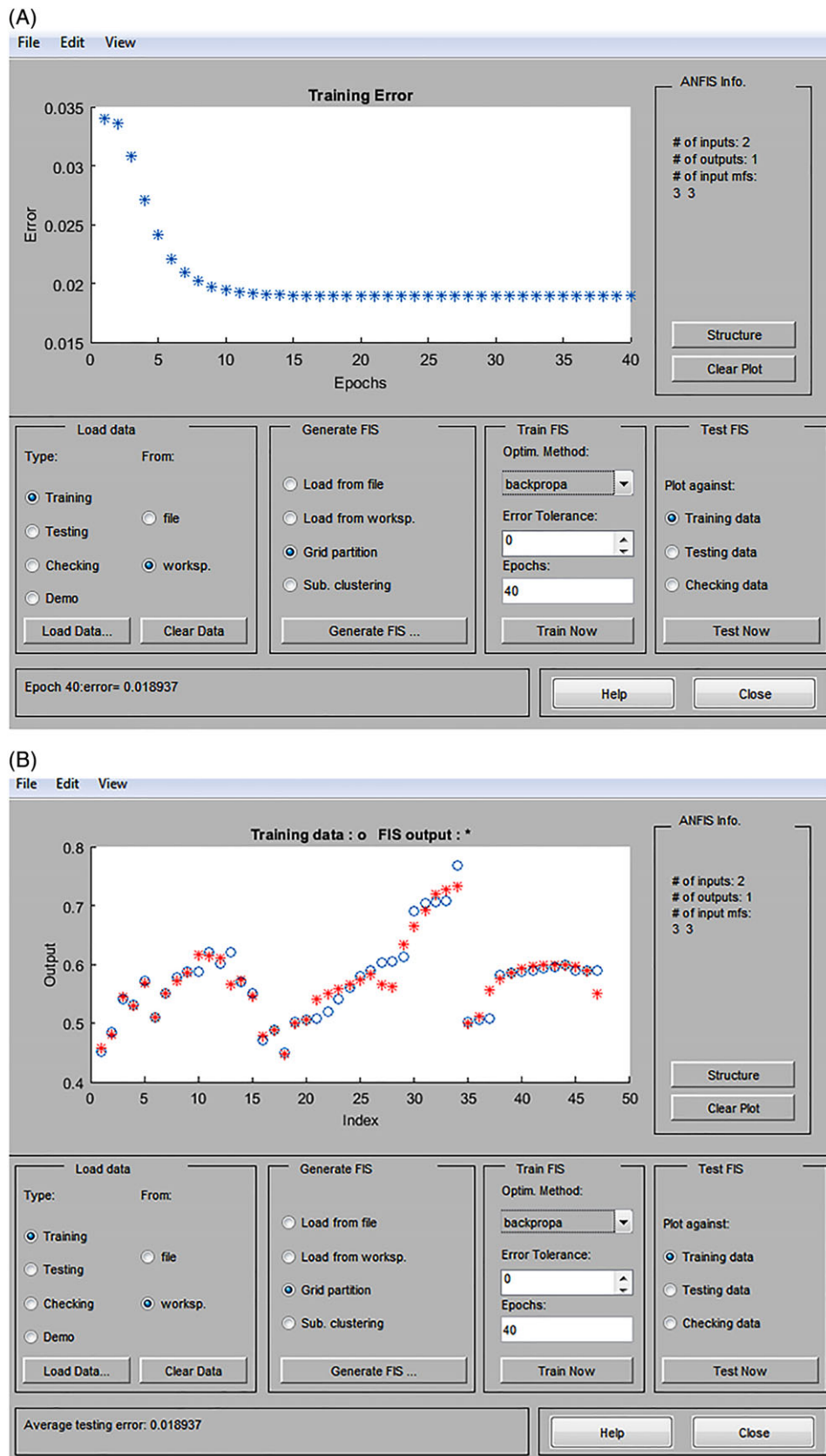


FIGURE 7 Continued.

$$o_i^3 = \bar{w}_i = \frac{w_i}{w_1 + w_2} \quad i = 1, 2, 3. \quad (3)$$

Layer 4: this layer is called as defuzzification layer. It calculates the individual output values “f” from the inferring of rule base. Each node in this layer is a square node, with the function as given by Equation 4 as

$$o_i^4 = \bar{w}_i f_i = \bar{w}_i (p_i x + q_i y + r_i), \quad (4)$$

where $i = 1, 2, 3$ and $\bar{w}_i$ is the output of layer 3 and $\{p_i, q_i, r_i\}$ is a parameter set. Parameters in this layer are also referred as consequent parameters.

Layer 5: this layer is called as the output layer. There is a single circle of the output neuron labeled as Σ , and the output signal is obtained by adding up input signals as it can be seen in the following Equation 5:

$$o_i^5 = \sum_i \bar{w}_i f_i = \frac{\sum_i \bar{w}_i f_i}{\Sigma} \quad (5)$$

where Equation 5 contains the components of the controller to regulate the DC bus voltage in the SAPF control strategy, and ANFIS-based MPPT algorithm controls the duty cycle of a boost converter to extract the maximum power. The ANFIS model for the controller design uses the architecture of the adaptive network. Figure 6 shows the flowchart for working operation of the ANFIS algorithm.

The ANFIS model is designed, trained, and analyzed using MATLAB 2015b Neuro-Fuzzy Designer tool-box and graphical user interface (GUI). A data set is collected using conventional methods as explained below and stored in the MATLAB workspace to train the ANFIS. The training data is different for SAPF controller and MPPT operation. The ANFIS is tested with different MFs such as triangular, trapezoidal, and Gauss MFs, and it is found that the training error is minimum with triangular MF. The epochs (iterations) and error for SAPF and MPPT controllers are shown in Figure 7.

2.3 | Data set for ANFIS-based SAPF controller

The conventional PI controller is designed and implemented to control the SAPF for varying load conditions and correspondingly the data set is recorded. This data set is stored in the MATLAB workspace and used to train the ANFIS network. The adaptive node parameters are updated using a learning algorithm to minimize error between target data and ANFIS output. ANFIS is given two inputs, ie, error and change in error. The output of the ANFIS controller is an active current component, which is used to maintain DC-link voltage and generates the required reference currents for the current controller to compensate the harmonic components. ANFIS-SAPF controller model has the following specifications: inputs, 2; output, 1; data pairs, 30; fuzzy rules, 9; nodes, 35; nonlinear parameters, 18; linear parameters, 27; total parameters, 45; and epochs, 40.

2.4 | Data set for ANFIS-based MPPT

Conventional P&O algorithm is used to track the MPP with varying temperature and irradiance, which in turn varies voltage and power. These are considered as inputs and the duty cycle of the boost converter is considered as an output. The duty cycle of DC-DC boost converter gets changed so as to receive MPP operation. This data set is recorded in the MATLAB workspace and is used to train the ANFIS network. The training was performed using back propagation algorithm. The ANFIS-based MPP controller model has the following specifications: inputs, 2; output, 1; training data pairs, 47; fuzzy rules, 9; nodes, 35; nonlinear parameters, 20; linear parameters, 27; total parameters, 47; and epochs, 40. The prediction error in the adaptive network can be minimized during training using hybrid network, which further optimizes the MF and its parameters.

By adjusting the values of MFs, ANFIS generates the set of fuzzy rules in order to produce appropriate output for different values of inputs. Parameters of MFs are adjusted or changed until the error gets reduced to a minimum value. Once all the parameters of MF are adjusted, the ANFIS model becomes a learning model ready to be used in a control scheme.

2.5 | Boost converter control to obtain MPPT

3 | CONTROL OF SHUNT ACTIVE POWER FILTER

3.1 | SRF theory-based ANFIS design

$$\begin{bmatrix} i_d \\ i_q \\ i_0 \end{bmatrix} = \frac{2}{3} \begin{bmatrix} \cos\theta & \cos\left(\theta - \frac{2\pi}{3}\right) & \cos\left(\theta + \frac{2\pi}{3}\right) \\ -\sin\theta & -\sin\left(\theta - \frac{2\pi}{3}\right) & -\sin\left(\theta + \frac{2\pi}{3}\right) \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} i_{La} \\ i_{Lb} \\ i_{Lc} \end{bmatrix}. \quad (6)$$

FIGURE 8 Block diagram of adaptive neuro-fuzzy inference system (ANFIS)-based control technique

$$i_d = i_{d-DC} + i_{d-AC} \quad (7)$$

$$= \frac{\sqrt{2}}{3} \left[i_{La} \cos(\omega t) + i_{Lb} \cos\left(\omega t - \frac{2\pi}{3}\right) + i_{Lc} \cos\left(\omega t + \frac{2\pi}{3}\right) \right]$$

$$i_q = i_{q-DC} + i_{q-AC} \quad (8)$$

$$= \frac{\sqrt{2}}{3} \left[-i_{La} \sin(\omega t) - i_{Lb} \sin\left(\omega t - \frac{2\pi}{3}\right) - i_{Lc} \sin\left(\omega t + \frac{2\pi}{3}\right) \right].$$

The $d - q$ to abc transformation is carried out to get three phase reference compensating currents i_{sa}^* , i_{sb}^* , i_{sc}^* and can be obtained as:

$$i_{sa}^* = \frac{\sqrt{2}}{3} [i_{d,AC} \cos(\omega t) - i_q \sin(\omega t)], \quad (9)$$

$$i_{sb}^* = \frac{\sqrt{2}}{3} \left[i_{d,AC} \cos\left(\omega t - \frac{2\pi}{3}\right) - i_q \sin\left(\omega t - \frac{2\pi}{3}\right) \right], \quad (10)$$

and

$$i_{sc}^* = \frac{\sqrt{2}}{3} \left[i_{d,AC} \cos\left(\omega t + \frac{2\pi}{3}\right) - i_q \sin\left(\omega t + \frac{2\pi}{3}\right) \right]. \quad (11)$$

The reference filter current is generated by comparing the reference source current and actual load current. The error generated by comparing the reference and actual filter currents is fed to hysteresis current controller, which generates the switching pulses for PV-SAPF.

4 | SIMULATION RESULTS AND DISCUSSION

The performance of combined PV-integrated SAPF system is evaluated using MATLAB/Simulink and then validated in real-time. The proposed system is evaluated on balanced and unbalanced nonlinear loading conditions. The simulation results are obtained under standard climatic conditions ($1,000 \text{ W/m}^2$ and 30°C). The results for various controller performances, SAPF actions, and THD are explained in the following sections.

4.1 | Transient response with various controllers

Figure 9 shows the output voltage of the combined (PV-SAPF) system with (1) ANFIS both in SAPF and MPPT, (2) ANN both in SAPF and MPPT, (3) FLC both in SAPF and MPPT, (4) PI in SAPF and P&O in MPPT, and (5) PI in SAPF

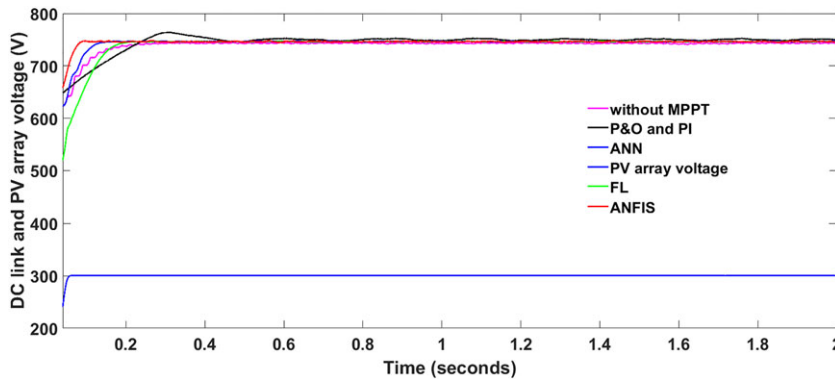


FIGURE 9 Transient response of DC-link voltage

and without MPPT. It also provides the PV array voltage without boost converter for reference. This Figure indicates that response with ANFIS-based strategy settles at 0.0989 s, ANN-based control technique settles at 0.1444 s, FLC-based algorithm settles at 0.2117 s, and P&O- and PI-based technique settles at 0.4484 s. Therefore, ANFIS-based control technique is faster than other techniques. It has lesser overshoots, and variation in DC-link voltage, ie, provides better transient response. Figure 10 shows the DC-link voltage under varying load condition.

4.2 | Performance of PV-SAPF without/with compensation

To demonstrate the effectiveness of turning on the SAPF, various current signatures are captured as in Figure 11. It can easily be observed that before turning on the SAPF, the load current (i_L) and source current (i_s) are highly distorted, and once the SAPF is turned on at 4.2 s, the i_s becomes purely sinusoidal. The further results with active SAPF are given in following sections.

4.3 | Performance of PV-SAPF with balanced and nonlinear load conditions

Figure 12 highlights the performance of the PV-SAPF when balanced and nonlinear load is applied. The source voltage (V_s) is 230 V and source current (i_s) is 158 A. The power factor obtained is 0.98. The load current (i_L) is 150 A, and the compensating current (i_c) is 37.5 A. Figures 13 and 14 show the harmonic spectrum of the source and load currents, respectively. The THD is reduced from 20.18% to 1.98%.

4.4 | Performance of PV-SAPF with unbalanced and nonlinear load conditions

Figure 15 shows the response analysis of PV-SAPF when nonlinear and unbalanced load is applied. The values of V_s and i_s are 230 V and 157 A, respectively. The power factor obtained is 0.99, which is close to unity. The values of i_L and i_c are 150 A and 37.4 A, respectively. Figures 16 and 17 show the harmonic spectrum of source and load currents, respectively. In this case, THD is reduced from 21.82% to 3.98%.

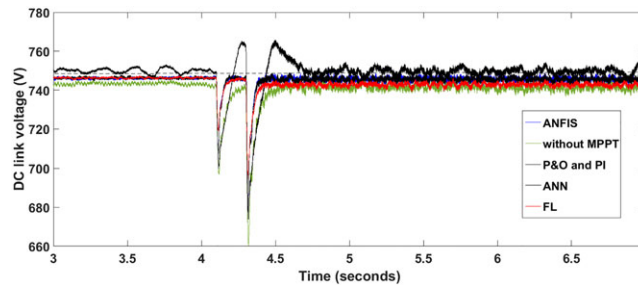


FIGURE 10 DC-link voltage under dynamically load changes

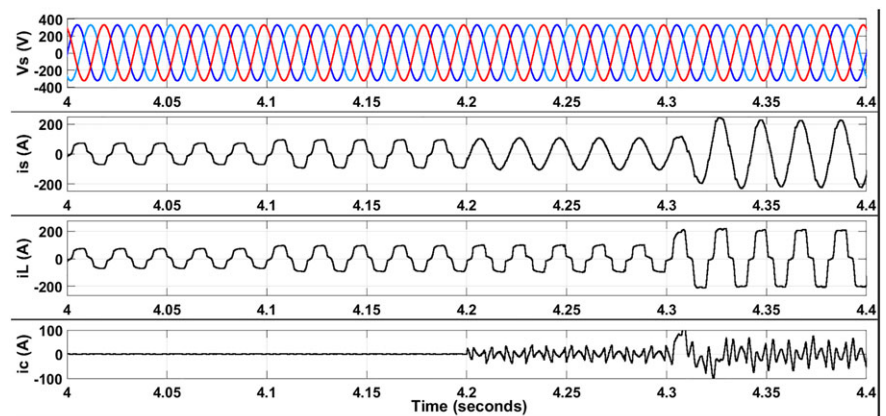


FIGURE 11 System performances without/with compensation

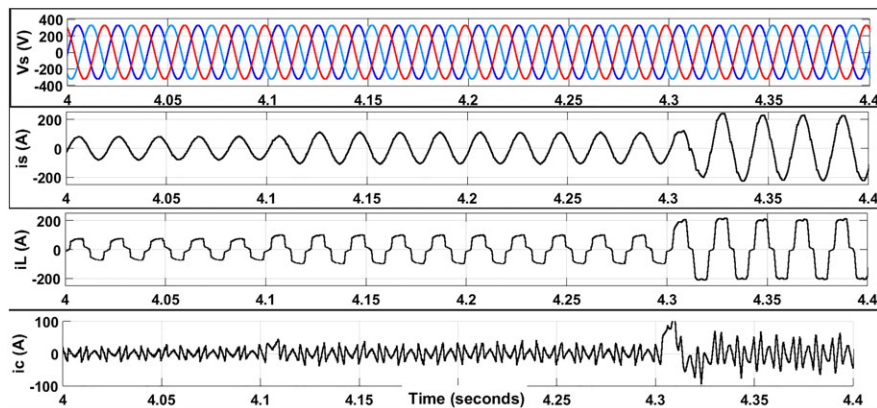


FIGURE 12 Performance of system with balanced and nonlinear load

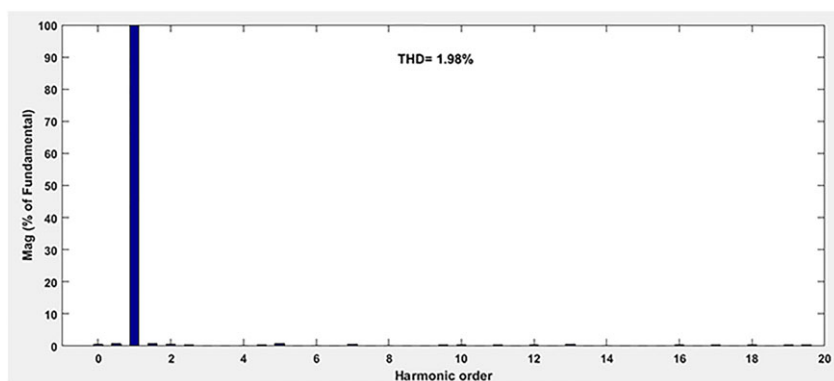


FIGURE 13 Harmonic spectrum of source current

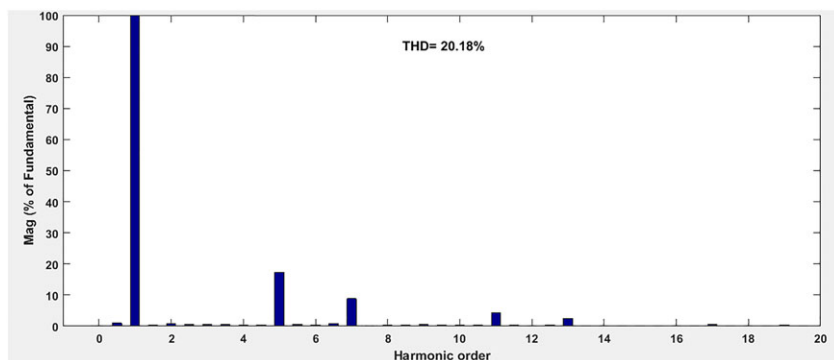


FIGURE 14 Harmonic spectrum of load current

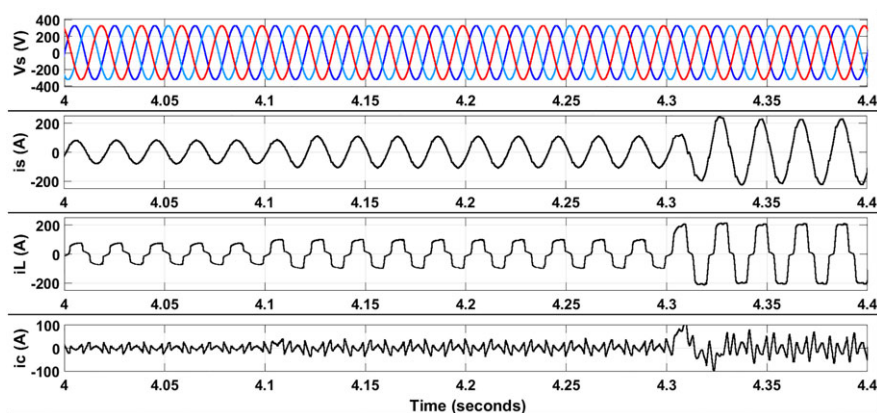


FIGURE 15 Dynamic performances of system with unbalanced and nonlinear load

FIGURE 16 Harmonic spectrum of source current

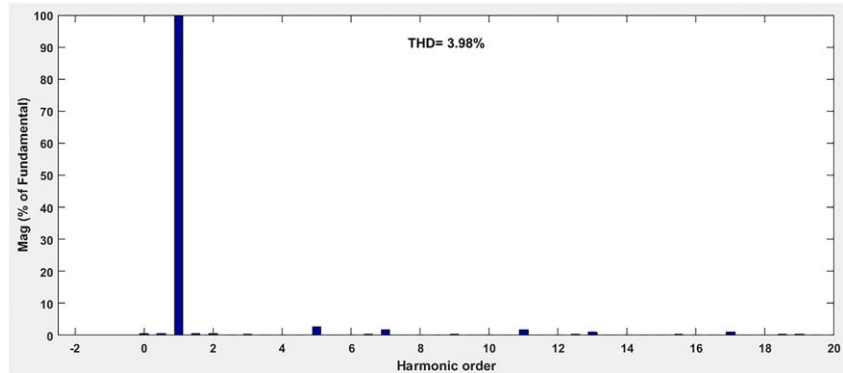
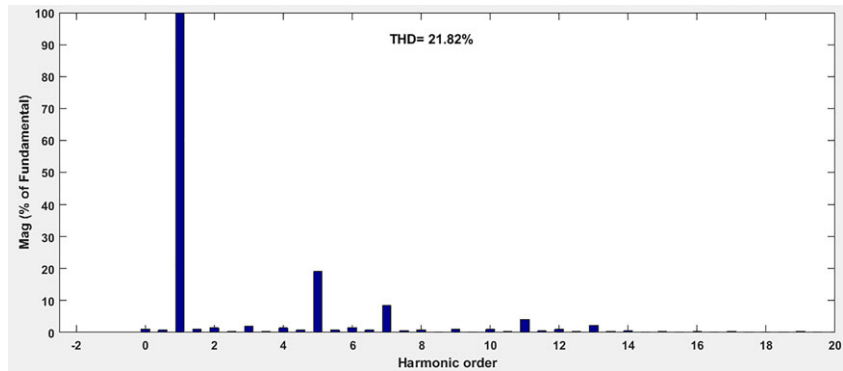


FIGURE 17 Harmonic spectrum of load current



5 | REAL-TIME RESULTS AND DISCUSSION

Real-time digital simulation hardware (RTDSH) technology is an essential part of the design and test procedure of different types of electric systems like power grids, adjustable speed drives, and power converters. RTDSH enables the user to readily convert Simulink models into real-time workshop and then to conduct real-time simulation on multiprocessor hardware design. It saves time, reduces risks for plant safety/operators, and offers many test functionalities.

In this paper, RT 4510 test system is used. This has kintex-7 FPGA-based high performance processor. The RTDSH output waveforms are observed with the help of a mixed signal oscilloscope (MSO). To accommodate the outputs, scale was decreased by a factor of 1,000 in voltage and current measurements. The RTDSH set up used to carry out the validation is given in Figure 18. Figure 19 characterizes the behaviors of DC-link and PV array voltages and their respective values are 780 V and 322 V.

5.1 | Performance of PV-SAPF with balanced and nonlinear load conditions

Figure 20 highlights the performance of the PV-SAPF when balanced and nonlinear load is applied. The figure depicts the signatures of V_s , i_s , i_L , and i_c . The values of V_s and i_s are 235 V and 162 A, respectively. The power factor obtained is 0.97. The values of i_L and i_c are 162 A and 37.4 A, respectively. Figure 21 shows the harmonic spectrum of load current. Figure 22 shows the harmonic spectrum of source current for ANFIS, ANN, and FLC strategies. It can easily be inferred that the THD using ANFIS method is minimum and its value is 0.97%.

5.2 | Performance of PV-SAPF with unbalanced and nonlinear load conditions

Figure 23 shows the response analysis for PV-SAPF when nonlinear and unbalanced load is applied. The values of V_s and i_s are 231 V and 167 A, respectively. The power factor obtained is 0.94. The values of i_L and i_c are 163 A and 34.4 A, respectively. Figure 24 shows the harmonic spectrum of load current. Figure 25 shows the harmonic spectrum of source current for ANFIS, ANN, and FLC strategies. The THD is reduced to 4.57% using ANFIS strategy.

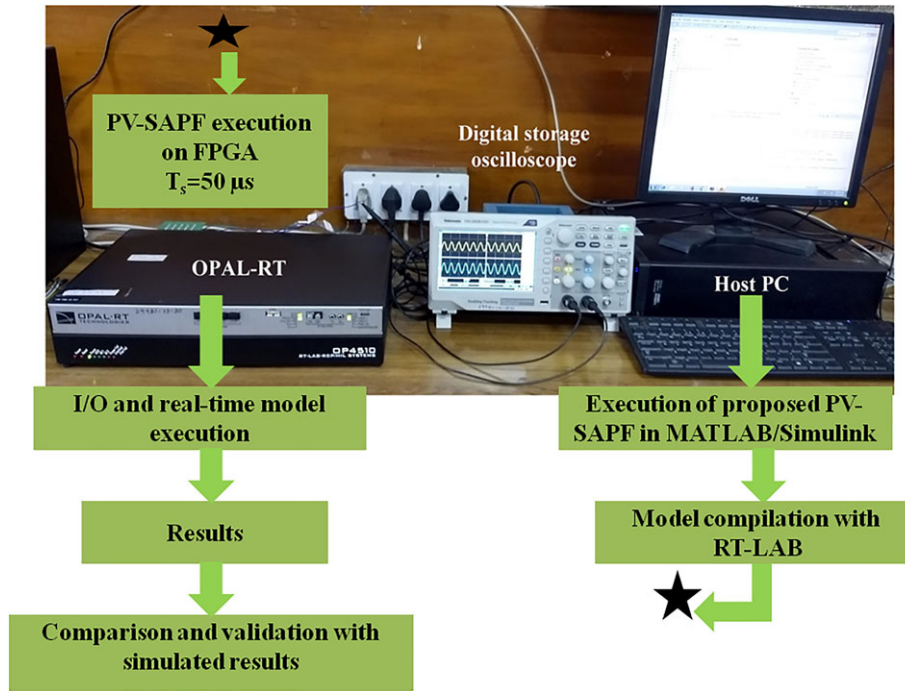


FIGURE 18 Real-time digital simulation hardware (RTDSH) laboratory set up

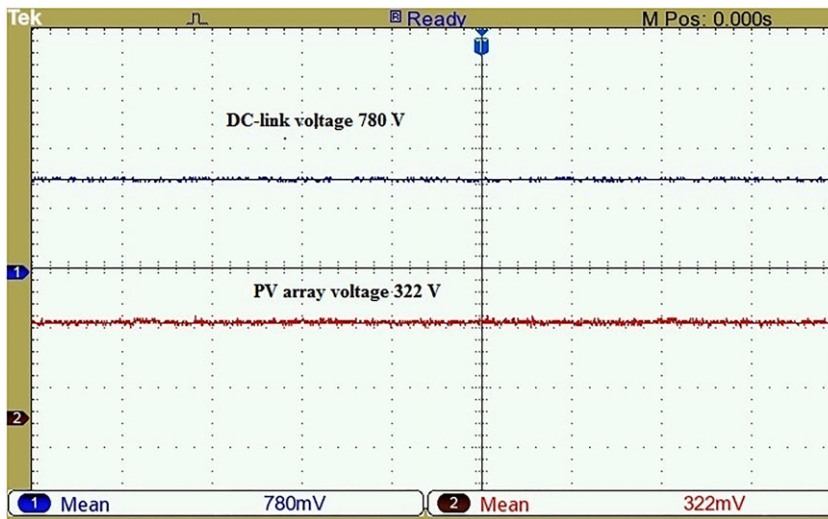


FIGURE 19 DC-link and photovoltaic (PV) array voltage

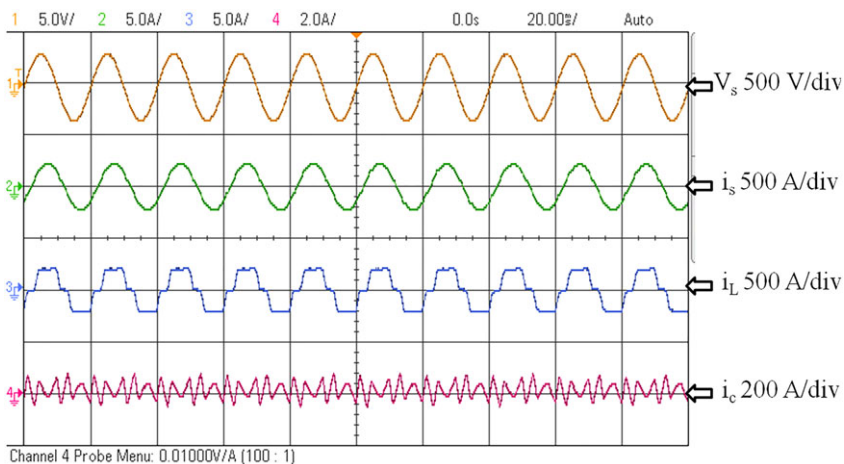


FIGURE 20 Performance of system with balanced and nonlinear load

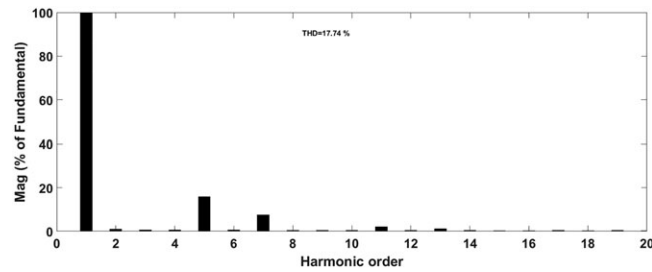


FIGURE 21 Harmonic spectrum of load current

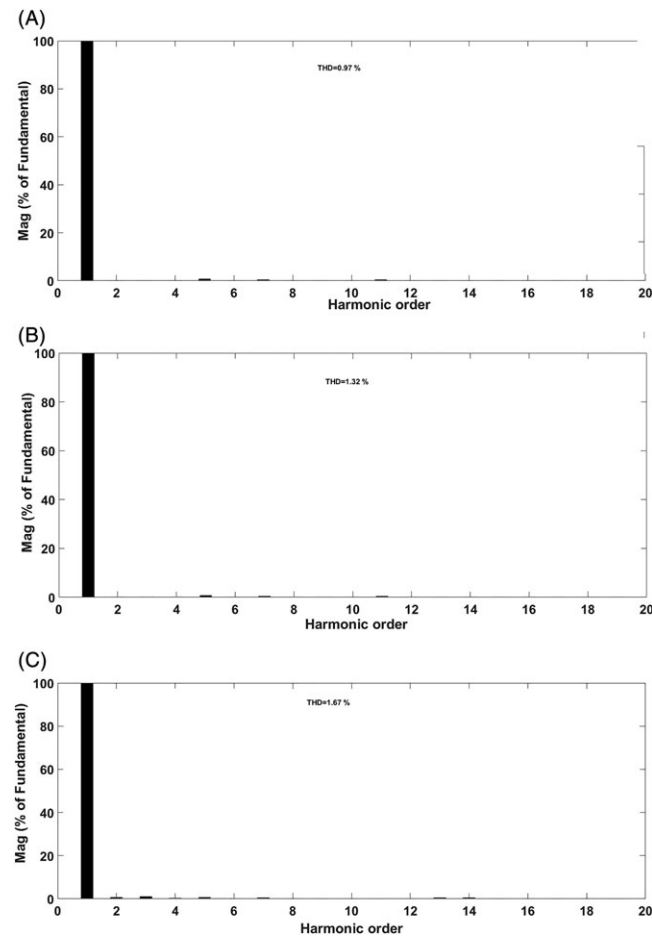


FIGURE 22 Harmonic spectrum of the source current using (A) adaptive neuro-fuzzy inference system (ANFIS) control technique, (B) artificial neural network (ANN) control technique, and (C) fuzzy control technique

5.3 | Comparison of PV-SAPF using different control technique

A comparative study of the proposed methods with traditional methods is carried out for balanced/unbalanced and nonlinear load conditions. Figures 26, 27, and 28 present the comparison in terms of THD, power factor, and DC-link voltage drop, respectively. The comparison shows that the ANFIS controller has better performance.

5.4 | Cost benefit analysis of harmonic reduction

The electronic equipment tends to be distributed in a premise on various branch circuits and most of the harmonic losses occur in its wiring. To evaluate the energy-loss impact of harmonics and reactive current flow, a wiring model was developed.³⁷ The harmonics result in the power losses and their cancellation leads to improved power quality and revenue saving. The PV-integrated SAPF can be suitably installed at selected points in a building/industrial

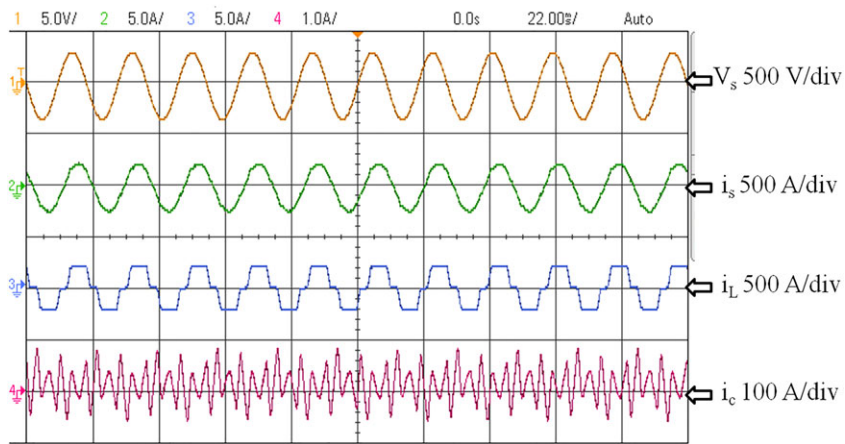


FIGURE 23 Performance of system under unbalanced and nonlinear load

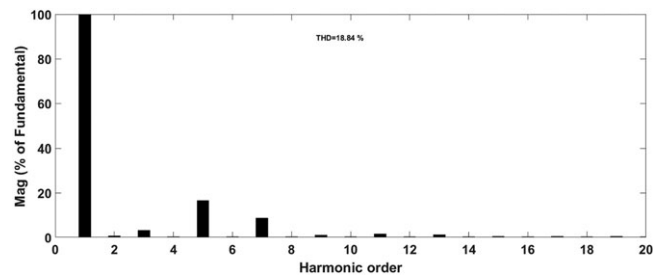


FIGURE 24 Harmonic spectrum of load current

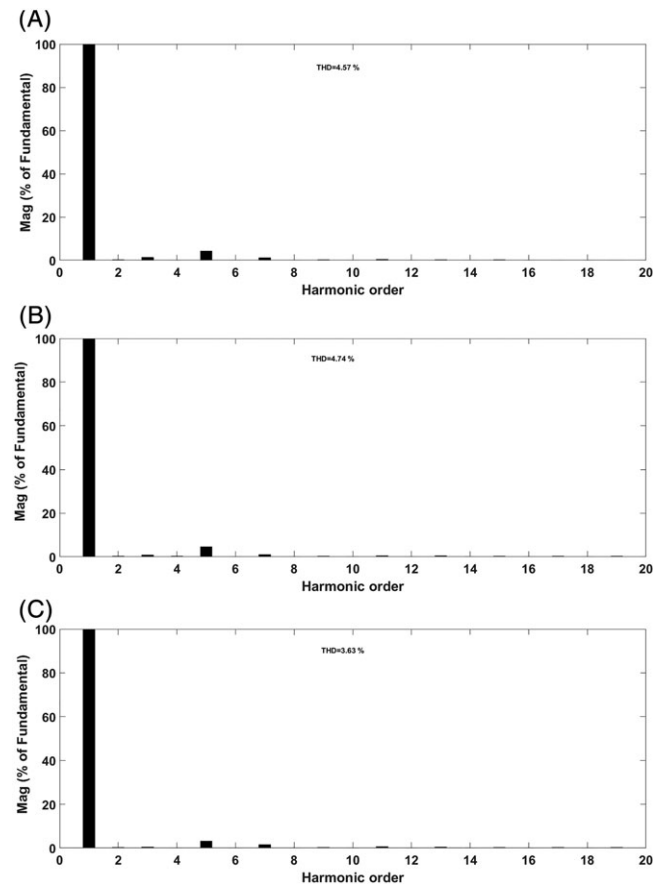


FIGURE 25 Harmonic spectrum of source current using (A) adaptive neuro-fuzzy inference system (ANFIS) control technique, (B) artificial neural network (ANN) control technique, and (C) fuzzy control technique

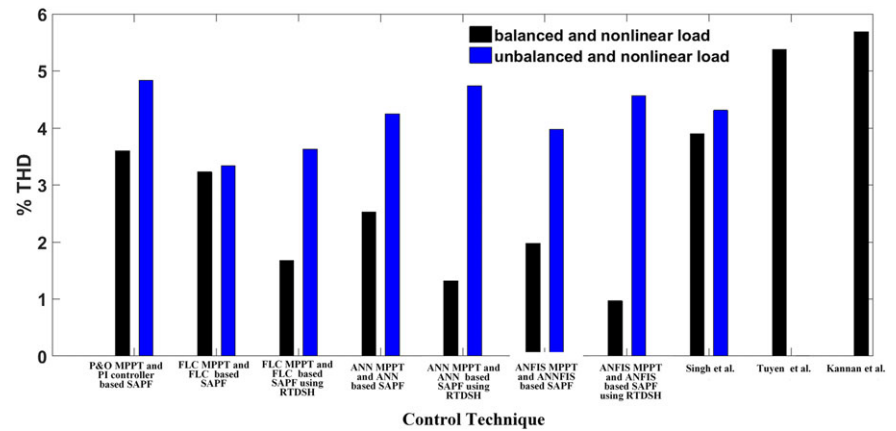


FIGURE 26 Comparison of source current total harmonic distortion (THD)

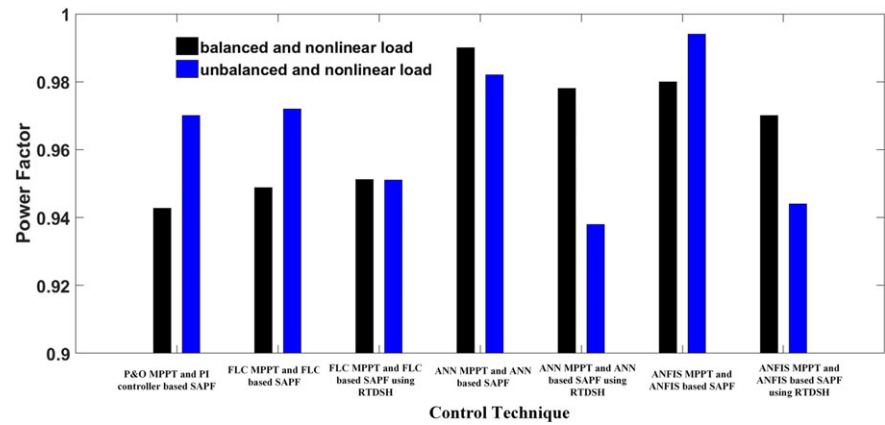


FIGURE 27 Comparison of power factor

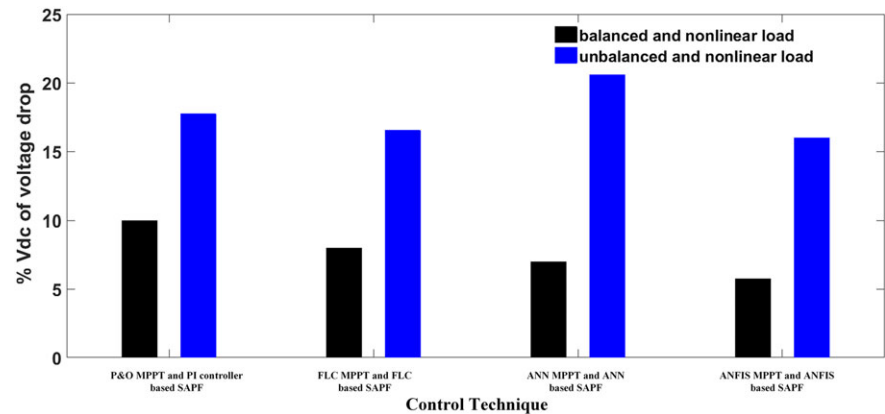


FIGURE 28 Comparison of DC-link voltage drop under dynamically load changes

plant/electrical distribution system to minimize the harmonics.³⁸ Energy savings can vary from 1% to 15%, depending on the level of harmonics. Harmonic suppression improves the power supply quality, and at the same time provides an energy saving benefit. The motivation of cost benefit analysis is received from the Key TS et al.³⁸

The losses and savings are compared for both compensated systems and non-compensated systems with different control techniques. The analysis given here is for a SAPF of 28 kVA and a load of 80 kW. The average cost of energy (in India) is taken here as ₹4/kWh, and the loading period is considered to be 10 hours. The losses and savings are compared for both compensated systems and noncompensated systems with different control techniques. Figure 26 illustrates the performance comparison of proposed system with other control techniques under balanced, unbalanced, and nonlinear load condition. The THD for PV-integrated SAPF system for balanced and nonlinear load condition using; ANFIS is 0.97%, ANN is 1.32%, and FLC is 1.67%, whereas for unbalanced and nonlinear load condition using ANFIS is 4.57%, ANN is 4.74%, and FLC is 3.63%. The THD for non-PV-integrated SAPF system using conventional

TABLE 2 Detailed comparison of control techniques

Control Technique	Balanced and Nonlinear Load			Unbalanced and Nonlinear Load		
	Total losses without compensation, W	Total losses with compensation, W	Saving per year, ₹	Total losses without compensation, W	Total losses with compensation, W	Saving per year, ₹
ANFIS	14 192	776	1,95,874	15 072	3656	1,66,674
ANN	14 192	1056	1,91,786	15 072	3792	1,64,688
FLC	14 192	1342	1,87,610	15 072	2904	1,77,653
Without PV (ANFIS)	14 192	2304	1,73,565	15 072	2944	1,77,069
Without PV (PID)	14 192	3656	1,53,825	15 072	4104	1,60,133

Abbreviations: ANFIS, adaptive neuro-fuzzy inference system; ANN, artificial neural network; FLC, fuzzy logic control; PID, proportional integral derivative; PV, photovoltaic.

control for balanced and nonlinear load condition using PID controller is 4.57%, whereas for unbalanced and nonlinear load condition, THD is 5.13%. These results are now compared in terms of cost incurred because of losses on account of harmonics and are shown in Table 2. It is observed that ANFIS-based control strategy provides minimum losses and maximum savings. The detailed comparison is given below.

Thus, PV-SAPF improves power quality, reduces energy costs, improves voltage conditions, eliminates power factor penalties, and advances system performance and productivity. In the above analysis, greater financial benefits may arise due to increased business productivity and improved product quality. On the basis of this analysis, the cost of PV-SAPF can be recovered in a period of 2 years. The results obtained from the proposed design are encouraging and can be of interest to industries, as well as commercial applications like aviation, textile, pharma, telecom, and metal, etc.

6 | CONCLUSION

To address the power quality issues imposed by EV charging, penetration of RES, and other nonlinear loads, a PV-integrated SAPF system with ANN-, FLC-, and ANFIS-based strategies are implemented in this paper. ANFIS, combining benefits of both ANN and FLC, yields better results. ANFIS-based PV-integrated SAPF not only improves the power quality but also supplies power to both grid and connected loads, as and when needed. The ANFIS-based PV-integrated SAPF provides THD well below 5%, which is within the IEEE-519 standard. The performance of this system is simulated in MATLAB/Simulink and then validated using RTDSH, OPAL-RT 4510.

The ANFIS is found superior in performance as it provides better control of reactive power compensation, reduces current harmonics, minimizes voltage variation as well as improves power factor and does not need an additional grid-tied inverter. The results obtained show that it has an excellent steady state and transient response for dynamic loading conditions. A cost-benefit analysis performed infers that the system is cost effective too. The simulated results are very close to the results obtained through RTDSH, thus it can be easily implemented in real situations. This system can be further tested and implemented on hybrid renewable energy sources to improve the power quality.

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How to cite this article: Kumar R, Bansal HO. Real-time implementation of adaptive PV-integrated SAPF to enhance power quality. *Int Trans Electr Energy Syst*. 2019;e12004. <https://doi.org/10.1002/2050-7038.12004>